Department of Information Technology Engineering

Title of the Project: “Raksha: Enhancing Women's Safety through AI-based Violence Detection.”

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Aim of the project

To develop an AI-powered safety system that uses real-time audio and video analysis to detect violent behavior and provide timely alerts, enhancing women’s safety through affordable and accessible technology.

Objectives

- Detect Violent Situations: Create an AI-based model capable of recognizing signs of violence based on audio (e.g., loud shouting) and visual cues (e.g., sudden movements).
- Real-Time Alerts: Develop a notification system to send alerts to emergency contacts or authorities when a violent situation is detected.
- Promote Safety: Provide a reliable solution to reduce response time and increase safety in areas with limited surveillance.

Applications

1) **AI-Powered Violence Detection:** Uses real-time audio and **v**ideo analysis to detect signs of aggression such as shouting, physical altercations, or sudden movements.

2) **Instant Emergency Alerts**: Automatically sends live alerts with location and evidence to emergency contacts or authorities upon detecting danger.

3) **Affordable Hardware Support**: Designed to run on basic cameras and microphones, making the system accessible for mobile phones and low-cost setups.

4) **Lightweight & Scalable Prototype:** Built as a resource-efficient model for small-scale testing and quick integration into homes, vehicles, or public spaces.

5) **Continuous Learning & Improvement**: Incorporates real-world testing data to refine detection accuracy and adapt to various environments and cultural contexts.

Results

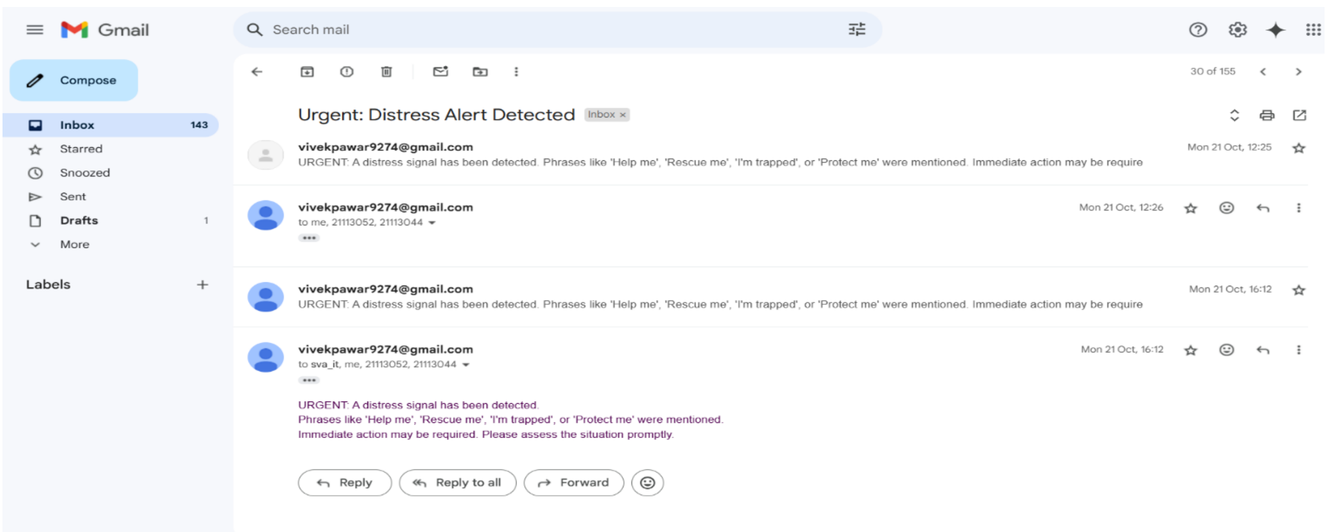
1) Created a working prototype capable of detecting violent behavior using **audio (shouting, distress sounds)** and **video (sudden/aggressive movements)** in real time

2) Developed and integrated a real-time alert system that notifies emergency contacts within seconds of detecting a threat.

3) Tested on **low-cost webcams and smartphone mics**, confirming system feasibility for use in **low-resource environments**.

4) Received encouraging feedback from initial user trials, highlighting **ease of use**, **quick alerts**, and **peace of mind** as key benefits.

Implementation Details



Conclusion

In conclusion, our presentation has provided a comprehensive overview of the project's objectives, system architecture, and proposed outcomes. We have discussed the technical requirements, tools, and methodologies employed to ensure a successful implementation. Moving forward, our focus will be on refining the system based on testing results and delivering a reliable, efficient solution within the project timeline.

Architecture Diagram